Plane Project Management - Deployment SOP

Document purpose: reproducible production deployment and operations guide for the Plane instance hosted on dte-apps.

Audience: the dte Linux deployment account and repository maintainers.

Production URL target: https://plane.pulsepharma.net:51026 over HTTPS.

> NOTE: Keep credentials out of this document, Git, terminal history, and chat. The

> NOTE: live server’s private environment files are the source of truth for secrets.

Application details

Item | Value

Application | Plane Project Management (Pulsepharma deployment)

VM hostname | dte-apps

VM public IP | 103.24.125.50 (confirm if the host IP changes)

VM user | dte

Application folder | /var/www/plane

GitHub organization | dte-pulse

GitHub repository | PlaneProjectManagment

Repository URL | https://github.com/dte-pulse/PlaneProjectManagment

Branch deployed | main

Runtime | Docker Compose production stack

Compose file | /var/www/plane/docker-compose.production.yml

Compose project | plane-pulsepharma

Private env directory | /etc/plane-pulsepharma

Systemd unit | plane-pulsepharma.service

Public hostname | plane.pulsepharma.net

Public access | HTTPS only; Sophos external port 51026 forwards to host port 443

TLS termination | Host Nginx, with the existing certificate for pulsepharma.net

Nginx upstream | 127.0.0.1:8106 by default

Database | External PostgreSQL on Aiven

Object storage | AWS S3 bucket hrappmodule (confirm bucket and region in private config)

Email | Microsoft 365 SMTP, STARTTLS on port 587

Deployment trigger | GitHub Actions push to main, or manual workflow dispatch

Actions workflow | .github/workflows/deploy-plane.yml

Service map

Plane area | Public path | Container/service

Main web app | / | web

API and authentication | /api, /auth, /static | api

Admin console | /god-mode | admin

Workspace/project UI | /spaces | space

Live collaboration | /live | live

Reverse proxy | Host loopback 8106; container port 80 | proxy

Background jobs | Internal only | worker, beat-worker

Cache and queue | Internal only | plane-redis, plane-mq

Sophos reserves its own public port 443, so Plane uses dedicated external HTTPS port 51026, forwarded to host port 443. The browser URL is therefore https://plane.pulsepharma.net:51026 unless the Sophos rule changes. The host may keep port 80 for other apps, but Nginx and Sophos must not route Plane traffic over HTTP. The Compose proxy binds only to loopback; its container port 80 is an internal upstream, not a public listener.

1. Prepare the host and network

This is a shared application host. Review existing listeners and services before changing firewall rules, Nginx, Docker, or systemd:

```
hostname
id dte
sudo ss -lntp
sudo systemctl status nginx docker --no-pager
```

Network requirements:

Traffic | Allow from | Purpose

External TCP 51022 | Administrator IPs or approved deployment source | Sophos SSH forward to host SSH

TCP 443 | Host interface | Plane HTTPS listener through Nginx

External TCP 51026 | Approved client networks | Sophos forwards to host TCP 443

External TCP 51020 | Other apps only, if required | Shared HTTP forward; must not route Plane

TCP 80 | Do not route to Plane; retain only for other host apps if needed | Other vhosts or certificate validation

Outbound TCP 443 | Required registries and AWS endpoints | Docker image/dependency downloads and S3

Outbound TCP 587 | Microsoft 365 SMTP endpoint | Email STARTTLS

Outbound PostgreSQL port | Aiven database endpoint; confirm provider port | Database traffic

Do not expose Compose service ports such as API 8000, web/admin/space 3000, RabbitMQ 5672, or Valkey 6379. The production Compose file exposes only the proxy on 127.0.0.1:${PLANEPROXYPORT}.

2. Confirm repository access and checkout

The host checkout must be owned by dte, and the host needs read access to the private repository. Use the host's configured GitHub deploy key/SSH alias if HTTPS authentication is unavailable. Do not put a personal access token in a Git URL.

For a new host:

```
sudo install -d -o dte -g dte -m 755 /var/www
sudo -u dte git clone --branch main \
  https://github.com/dte-pulse/Plane_Project_Managment /var/www/plane
```

For the existing host:

```
cd /var/www/plane
git status --short --branch
```

```
git remote -v
git branch --show-current
```

Do not overwrite local files or reset the checkout if it contains uncommitted work. The current deployment workflow expects /var/www/plane and branch main.

3. Configure private runtime files

Production secrets belong outside the repository in /etc/plane-pulsepharma. The expected files are:

File | Purpose

compose.env | Public URL, proxy port, env-file paths, S3 bucket, trusted proxy ranges

api.env | Aiven URL, Django key, AWS credentials, SMTP, service URLs, app settings

live.env | Live collaboration service URLs and matching live secret

root.env | RabbitMQ container credentials

Caddyfile.external | Internal Plane proxy routing configuration

For a new installation, create the directory and copy examples, then set permissions and edit values locally on the host:

```
sudo install -d -o dte -g dte -m 700 /etc/plane-pulsepharma
cd /var/www/plane
sudo install -o dte -g dte -m 600 deployments/plane/production.env.example /etc/plane-pulsepharm
a/compose.env
sudo install -o dte -g dte -m 600 deployments/plane/api.env.example /etc/plane-pulsepharma/api.e
nv
sudo install -o dte -g dte -m 600 deployments/plane/live.env.example /etc/plane-pulsepharma/live
.env
sudo install -o dte -g dte -m 600 deployments/plane/root.env.example /etc/plane-pulsepharma/root
.env
sudo install -o dte -g dte -m 600 deployments/plane/Caddyfile.external /etc/plane-pulsepharma/Ca
ddyfile.external
```

When maintaining an existing deployment, edit the private files in place. Do not copy example files over production files: that would replace credentials and deployment-specific values.

compose.env

Required target values:

```
PLANE_PUBLIC_URL=https://plane.pulsepharma.net:51026
PLANE_PROXY_PORT=8106
PLANE_ROOT_ENV_FILE=/etc/plane-pulsepharma/root.env
PLANE_API_ENV_FILE=/etc/plane-pulsepharma/api.env
PLANE_LIVE_ENV_FILE=/etc/plane-pulsepharma/live.env
PLANE_PROXY_CONFIG=/etc/plane-pulsepharma/Caddyfile.external
AWS_S3_BUCKET_NAME=hrappmodule
TRUSTED_PROXIES=<nginx-and-docker-proxy-cidrs>
```

PLANEPUBLICURL has no path or trailing slash. With the current Sophos mapping, use https://plane.pulsepharma.net:51026; that external port forwards to host port 443. The host proxy port is loopback-only and is not user-facing.

Application secrets and service settings

Set production values in api.env, live.env, and root.env; do not copy the placeholder values verbatim.

- Aiven PostgreSQL: use the provider URL and SSL mode required by the

database. URL-encode reserved characters in the password.

- Django SECRETKEY: preserve the currently deployed key when connecting to an existing Plane database. Plane uses it to decrypt saved integration values. Changing it can make stored encrypted values unreadable.

- AWS S3: use a least-privilege IAM credential scoped to the required bucket. Keep USEMINIO=0; omit AWSS3ENDPOINTURL for AWS S3.

- Redis/Valkey: REDISHOST=plane-redis, port 6379, URL redis://plane-redis:6379/.

- RabbitMQ: host plane-mq, port 5672, vhost plane; credentials in api.env must match root.env.

- Service origins: CORSALLOWEDORIGINS, WEBURL, APPBASEURL, ADMINBASEURL, SPACEBASEURL, and LIVEBASEURL must use https://plane.pulsepharma.net:51026. Keep configured subpaths such as /god-mode, /spaces, and /live in their respective BASEPATH settings.

- Live secret: LIVESERVERSECRETKEY must match exactly between api.env and live.env.

- SMTP: Microsoft 365 uses smtp.office365.com, port 587, TLS enabled, SSL disabled. Keep the mailbox app password only in api.env.

- S3 bucket consistency: AWSS3BUCKETNAME in compose.env and api.env must agree.

Generate random secrets on the host without entering them in shell history:

```
openssl rand -hex 32
```

Review permissions without displaying file contents:

```
sudo ls -ld /etc/plane-pulsepharma
sudo ls -l /etc/plane-pulsepharma
```

4. Validate and build the production stack

Run commands from the repository root, always using the deployment project name and external env file:

```
cd /var/www/plane
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml config --quiet
```

Back up the Aiven database before migrations. Do not run two independent Plane versions against the same live database. If local development services are running against the production database, stop them before production migrations. Build the images:

```
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml build
```

Start the internal data services, run migrations, then start Plane:

```
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml \
  up -d plane-redis plane-mq

docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml run --rm migrator
```

```
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml \
  up -d web admin space api worker beat-worker live proxy
```

The application services are web, admin (/god-mode), space (/spaces), API, worker, beat-worker, live collaboration, and the internal proxy. The database is external; production Compose does not create a local PostgreSQL service.

SMTP database synchronization

Plane stores SMTP configuration in the database. After API startup, apply the private api.env SMTP values using the repository helper. It encrypts the password with Django SECRETKEY and does not print it:

```
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml \
  exec -T api python manage.py shell < deployments/plane/configure-smtp.py
```

5. Configure HTTPS-only Nginx

Create/verify the DNS A record plane.pulsepharma.net pointing to 103.24.125.50. The current Sophos routes use external SSH port 51022 and shared HTTP port 51020; Plane uses its own external HTTPS port 51026, which forwards to host Nginx port 443. Sophos itself owns external port 443. The user-facing URL is https://plane.pulsepharma.net:51026.

Nginx terminates TLS on host port 443. Install the repository's HTTPS-only example as a separate virtual host, then adapt the certificate paths if needed:

```
sudo cp /var/www/plane/deployments/plane/nginx.plane.conf.example \
  /etc/nginx/sites-available/plane.pulsepharma.net
sudoedit /etc/nginx/sites-available/plane.pulsepharma.net
sudo ln -s /etc/nginx/sites-available/plane.pulsepharma.net \
  /etc/nginx/sites-enabled/plane.pulsepharma.net
```

The server block must listen on 443 ssl, use the valid wildcard certificate, proxy to http://127.0.0.1:8106, and set X-Forwarded-Proto https. Do not create or enable a Plane HTTP server block on port 80. On an existing server, locate any old Plane HTTP site and remove/disable only that Plane site; preserve other applications' Nginx virtual hosts.

Validate before applying Nginx changes:

```
sudo nginx -t
sudo systemctl reload nginx
sudo systemctl status nginx --no-pager
```

In Sophos/firewall/NAT, forward external TCP 51026 to this host's TCP 443 and allow only approved client networks where applicable. Sophos reserves external port 443. Do not forward HTTP port 51020 or host port 80 to Plane. If another app requires HTTP on the shared host, its port 80 rule may remain, but Plane must not answer or redirect from it.

6. Install the systemd unit

The repository unit is configured for user/group dte, working directory /var/www/plane, and private compose.env:

```
sudo cp /var/www/plane/deployments/plane/plane-pulsepharma.service.example \
  /etc/systemd/system/plane-pulsepharma.service
sudo systemctl daemon-reload
sudo systemctl enable --now plane-pulsepharma.service
```

```
sudo systemctl status plane-pulsepharma.service --no-pager
```

The unit is Type=oneshot with RemainAfterExit=yes; active (exited) is normal. Docker's unless-stopped policy keeps containers running and restarts them after host boot. The unit is scoped to Compose project plane-pulsepharma; do not use an unscoped docker compose down on this shared host.

7. Configure GitHub Actions deployment

The workflow at .github/workflows/deploy-plane.yml runs on pushes to main and also supports manual dispatch from GitHub Actions. Configure these repository secrets under Settings -> Secrets and variables -> Actions:

Secret | Value

EC2HOST | SSH-reachable hostname or IP for dte-apps

EC2USER | dte

EC2PORT | 51022 for the current Sophos SSH forward (confirm if changed)

EC2SSHKEY | Private SSH key whose public key is authorized for dte

The Actions key is the VM-login key; it is separate from any GitHub deploy key that the VM uses to pull the private repository. Restrict the key to this host account. The deployment account must be able to pull main, run Docker Compose, and read all private env files. The current workflow does not use sudo.

For each deploy, Actions:

1. SSHes to the host and enters /var/www/plane.
2. Pulls origin main with fast-forward only.
3. Validates production Compose without printing config values.
4. Builds production images.
5. Runs the database migrator.
6. Starts/recreates the Plane Compose services and reports compose ps.

The workflow serializes production deployments. It does not run docker compose down, delete volumes, or reload Nginx. A push that changes frontend origins requires the server's private compose.env to have the new origin before the build so those values are embedded in the frontend images.

8. Verify the deployment

On the host, check service state and the private upstream:

```
cd /var/www/plane
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml ps
curl -I http://127.0.0.1:8106/
sudo nginx -t
sudo systemctl is-active plane-pulsepharma.service
```

From an approved client network:

```
curl -I https://plane.pulsepharma.net:51026/
```

The browser URL is https://plane.pulsepharma.net:51026. Confirm the TLS certificate is valid and that login, workspace pages, admin, file uploads, email, and live collaboration work. Confirm Plane is not reachable over HTTP on port 80. Do not infer that from the Compose port mapping alone; check the active Nginx site and upstream firewall.

9. Routine deployment and rollback

Normal releases go to main; GitHub Actions performs the deployment. If deploying manually, use the same ordered steps: fast-forward pull, Compose validation, build, database backup when applicable, migration, then up -d for the application services. Avoid restarting unrelated host applications.

If a code release fails:

1. Read the failed Actions log and host container logs.
2. Identify the last known-good commit and check whether the failed release introduced database migrations.
3. Revert the code change with a new commit or deploy a known-good commit using the repository's approved release process.
4. Rebuild and start the services; do not delete Compose volumes.
5. Restore a database backup only when the migration/data change requires it and the restore plan has been reviewed. A code rollback alone may not reverse database migrations.

Never use docker compose down -v for routine deployment or rollback; it deletes Plane's local RabbitMQ and Valkey data volumes.

10. Troubleshooting

Symptom	Check
Actions SSH connection refused	Confirm EC2HOST, EC2PORT, firewall/NAT SSH forwarding, and host SSH service.
SSH authentication failure	Confirm EC2SSHKEY matches the public key in dte's authorizedkeys.
git pull fails	Check the host deploy key/SSH alias and read access to the private repository.
Compose says a variable is missing	Check compose.env path and required PLANE values; keep private values out of logs.
Image build fails	Inspect the first failing Docker build step; warnings about large bundles or slow registry requests are not necessarily fatal.
Migration fails	Check Aiven reachability, database URL/SSL mode, permissions, and backup status before retrying.
Site unavailable on HTTPS	Check container health/logs, loopback proxy, active Nginx site, certificate paths, and firewall 443 forwarding.
Site still loads over HTTP	Find and disable the Plane-specific Nginx port 80 site and remove any firewall/NAT route to Plane. Do not disable shared HTTP sites without checking their owners.
Email does not send	Confirm private SMTP settings and rerun configure-smtp.py after API startup.
Collaboration disconnects	Confirm /live routing, WebSocket headers, LIVESERVERSECRETKEY equality, and origins.
S3 uploads fail	Verify bucket, region, IAM permissions, and matching bucket settings; never print AWS keys.

View logs without displaying environment files:

```
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f docker-compose.production.yml \
  logs --tail=100 api worker beat-worker live proxy web admin
sudo journalctl -u plane-pulsepharma.service -n 100 --no-pager
```

11. Operational safety and backups

- Back up the external Aiven PostgreSQL database before migrations and confirm the recovery procedure periodically.
- Keep separate least-privilege AWS and database credentials where possible.
- Rotate any credential that has been exposed and update the corresponding

private env file; never commit secrets.

- Preserve Django SECRETKEY when reusing the existing database.

- Preserve RabbitMQ and Valkey Compose volumes. Do not prune volumes as part of deploy cleanup.

- On this shared server, keep all Compose commands explicitly scoped with -p plane-pulsepharma and the production Compose file.

- Keep public access HTTPS-only. Sophos external port 51026 forwards to host port 443; the internal proxy port 80 is loopback-only and not public.

Quick reference

```
# Status
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f /var/www/plane/docker-compose.production.yml ps

# Logs
docker compose --env-file /etc/plane-pulsepharma/compose.env \
  -p plane-pulsepharma -f /var/www/plane/docker-compose.production.yml \
  logs --tail=100 api worker beat-worker live proxy

# Nginx validation/reload after a config change
sudo nginx -t && sudo systemctl reload nginx

# systemd status
sudo systemctl status plane-pulsepharma.service --no-pager
```